AUB Course Planner: An MCP Toolkit

C5 Agent Tools: Part 2 - Reflection

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Fall 2026–2027

Reflection

I drew the line between the tools by asking which tool owns each rule, using the "and" test from class. `get_course_info` only reads a course, `add_course_to_schedule` only registers one section of one course, and `plan_next_semester` only decides what to try based on the student's preferences. The key decision was to keep every AUB registration rule (prerequisites, time conflicts, seats, credit limit) inside the Action tool alone. The Orchestration tool never re-checks them: it calls the Action tool section by section and reacts to its errors, which is how it works around full sections and time conflicts. This kept each rule in one place, so testing or fixing a rule never meant changing two tools. It also made the system safe against tricks: when I told the agent to ignore its instructions or claimed an approval that was not in the record, registration still went through the Action tool, which checked the real record, and no tool can change grades at all. Read-only data the agent always needs, like the student's record, became MCP resources instead of extra tools. Because the Agents SDK does not pass resources to the model automatically, the client reads the profile at startup and adds it to the agent's instructions.

Three tools felt right for the core flow of looking up, registering and planning, and the limit pushed me to keep each one focused instead of building one large planner tool. The limit did show in one place: there is no way to drop a course, so the schedule is reset with a script, and adding drop to the Action tool would have broken the single-purpose rule. The biggest surprise was that the model sometimes tried to do the tools' job, refusing courses from the prerequisite text without calling the Action tool; clear instructions to always let the tool decide fixed this. Next steps would be a `drop_course` Action tool, real section data from the registrar instead of sample sections, time preferences (for example, no 8 AM classes) so the planner can choose between valid sections, and planning across several semesters toward graduation.